# Aman Verma

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## WORK EXPERIENCE

**Software Engineer Intern** Nov 2025 - July 2026

*LexoLegal* *Remote*

- Architected a cross-platform case management client (Android/iOS) in **Kotlin Multiplatform (KMP)** and **Compose Multiplatform**, structuring an adaptive shell on **Jetpack Navigation 3**, **Koin DI**, and **Room KMP** with **85%+ code sharing** (**168 unit tests**).
- Engineered an automated 12-hour case refresh sweep in **FastAPI**, **Celery**, and **Redis** with an idempotent diffing engine, delivering real-time push alerts via **Firebase Cloud Messaging** across **25 High Courts** with **zero duplicate alerts**.
- Deployed 'Munshi', an autonomous legal assistant in **Pydantic AI** and **FastAPI** interfacing **Model Context Protocol (MCP)** servers, automating legal brief retrieval and hearing scheduling with **sub-2s latency** across **38 REST endpoints** (**1,024 pytests**).

**Freelance Developer** Oct 2024 - Jan 2025

*Webxela* *Remote*

- Shipped Coupit, a cross-platform POS loyalty client in **Compose Multiplatform (CMP)**, engineering a **60 FPS** physics-driven Canvas spin wheel, **Koin DI**, and dynamic on-device QR voucher generation across **Android and iOS**.
- Integrated Square POS **OAuth 2.0** and webhook ingestion in **Spring Boot** and **PostgreSQL** (**7 relational entities**), automating merchant onboarding in **under 2 minutes** with dynamic **Apple Wallet** pass generation via **PassKit**.
- Developed a concurrent trademark search engine in **Kotlin** and **Spring Boot**, executing **25 parallel coroutine worker threads** with connection pooling and **Flyway** batch persistence to sweep **5,000-record chunks** across public registries.

## PROJECTS

**WatchTower** | *Kotlin, Jetpack Compose, WorkManager, Coroutines* GitHub

- Developed a local-first Android web monitoring app utilizing MD5 checksum hashing and **Myers diff algorithm** to track DOM modifications and dispatch real-time push alerts.
- Optimized concurrent network inspection via **Kotlin Coroutines** with semaphore rate limiting, executing background crawls via **WorkManager** while maintaining a consistent **60 FPS UI frame rate**.
- Constructed persistent storage using **Room Database** with **Flow** observables and integrated **OkHttp** client with custom SSL trust managers, rendering visual diffs with bounded memory (**<150MB**).

**Oak** | *Kotlin Multiplatform (KMP), Compose Multiplatform, Ktor, Koin* GitHub

- Authored an open-source **Kotlin Multiplatform** library targeting **Android, iOS, and JVM** to extract and normalize catalog and pricing data across **5+ major e-commerce platforms**.
- Structured **Clean Architecture** with platform-agnostic domain use cases, multi-engine **Ktor** HTTP client abstractions, and custom **Gradle convention plugins** for automated publication.
- Abstracted multi-platform e-commerce pagination and product schema mappers using **Koin DI**, decoupling network scrapers to achieve **100% shared business logic** across consumer client apps.

**Cypher-X** | *Kotlin, Jetpack Compose, AndroidKeyStore, Biometrics, Room* GitHub

- Built an offline-first encrypted password manager utilizing hardware-backed **AndroidKeyStore** and **AES-256-GCM** authenticated encryption with cryptographically random initialization vectors.
- Implemented biometric authentication gates using **AndroidX BiometricPrompt** and **MVI** unidirectional data flow to secure local **Room** database persistence against unauthorized access.
- Engineered a modular Clean Architecture domain layer with **Koin DI** and reactive **Flow** pipelines, managing secure credential lifecycle states with **100% local-first storage** in **Room**.

## TECHNICAL SKILLS

**Languages & Core**: Kotlin, Python, SQL, DSA, OOP, DBMS, Operating Systems

**Mobile & Multiplatform**: Android SDK, Jetpack Compose, Kotlin Multiplatform (KMP), Compose Multiplatform (CMP), Retrofit, Room, SQLite, WorkManager, Jetpack Navigation

**Backend & Cloud**: Spring Boot, Spring Data JPA, Spring Security, Ktor, FastAPI, Celery, Redis, PostgreSQL, REST APIs, WebSockets, Firebase

**Architecture & Async**: Clean Architecture, MVI, MVVM, Multi-Module Architecture, Kotlin Coroutines, Flow, StateFlow, Multithreading, Dependency Injection, Hilt, Koin

**Testing & Tooling**: JUnit 5, MockK, Kotest, Gradle (Kotlin DSL), Git, GitHub Actions, Docker, Android Studio, Postman

## LEADERSHIP & ACTIVITIES

- Delivered 3 technical sessions as a speaker on 'Generative AI Unlocked: LLM Basics' and 'Level Up Your Career with AI', training 150+ college students on foundational LLM concepts, prompt engineering, and modern AI developer tools.
- Coordinated placement drives and corporate outreach as Student Coordinator at CTPC (Central Training & Placement Cell), managing logistics and interview scheduling for recruiters and 100+ candidates.
- Organized Smart India Hackathon internal rounds across 2 editions; coordinated problem statement curation, mentoring, and judging logistics for 20+ teams and 120+ participants.

## EDUCATION

**Veer Bahadur Singh Purvanchal University** 2023 - 2027

*Bachelor of Technology in Computer Science & Engineering* *Jaunpur, India*